

Software Requirements Specification (SRS)

Project: Online Internship & Placement Preparation Portal

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Vyshali Aytha, Vismaya Harish, Thejus A Reddy, Yashas Sadananda	Initial SRS draft	

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Online Internship & Placement Preparation Portal, a college mini-project. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web-based platform that helps students discover internship and placement opportunities and prepare for them, while giving recruiters and the institution's placement cell simple tools to manage the process.

1.2 Scope

Covers student registration and profile management, resume upload/download, browsing and applying to internship/placement listings, mock tests with auto-scoring, an interview-question bank with mock-interview slot booking, recruiter job postings and candidate shortlisting, and placement-cell approval/reporting. Excludes payroll/HR onboarding of recruiting companies and any activity after a candidate formally joins an organisation.

Not part of the current implementation (listed for context; may be considered as future/optional enhancements beyond this mini-project): institutional single-sign-on (SSO), automated resume parsing, integrated video-conferencing for mock interviews, payment-based premium test packs, and large-scale (thousands of concurrent users) load handling.

1.3 Audience

Students, Recruiters, Placement Cell Administrators, Developers, and Evaluators.

1.4 Definitions

List of acronyms: SRS, UI, API, OTP, JWT, CGPA.

2. Overall description

2.1 Product perspective

The Portal is a web application, accessed by students, recruiters, and placement-cell staff through a browser, backed by a relational database. It uses a basic email/SMS gateway for OTP delivery; no other third-party service integration is required for the current scope.

2.2 Major product functions (detailed)

- Student registration, authentication and profile management
- Resume upload and download
- Internship and placement listing search and filters
- Application submission and status tracking
- Mock tests with auto-scoring
- Interview question bank and mock-interview slot booking
- Recruiter job posting, applicant viewing and shortlisting
- Placement-cell approvals and basic reporting

2.3 User roles and characteristics (expanded)

- Student: basic computer literacy, expects a simple application flow and timely status updates.
- Recruiter: needs a simple way to post listings and shortlist applicants with minimal training.
- Placement Cell Admin: approves recruiter/company registrations and generates placement reports.

2.4 Operating environment

Modern web browsers (Chrome, Firefox, Edge) on desktop and mobile; backend hosted on a single application server with a relational database, suitable for a college-scale deployment.

2.5 Constraints

Student personal data must be handled responsibly; the system is built and evaluated within a single academic term by a small student team, so third-party integrations are kept to a minimum (email/SMS only).

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web dashboard with role-based views (Student, Recruiter, Admin). Accessibility: clear labels and adequate colour contrast.

3.2 Hardware interfaces

Standard desktop/laptop/mobile devices with a browser; no specialised hardware required.

3.3 Software interfaces

- Email/SMS Gateway API for OTP delivery.

3.4 Communications

- HTTPS (TLS) shall be used for all client-server communication.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow OIPP-F-###.

4.1 Registration & Authentication

Description: Each role has a distinct onboarding path. Students self-register directly. Recruiters submit an account request that stays pending until the placement-cell admin approves it (see 4.8). Admin accounts are pre-created by the system/institution and are not self-registered. All approved users then sign in with email and password.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-001	The system shall allow a student to self-register using their institutional email and verify the account via an OTP sent by email.	Functional	High	Student	AC-OIPP-F-001: Registration succeeds only after correct OTP entry. Test: TC-Auth-01	Requires Email/SMS gateway
OIPP-F-002	The system shall allow any approved user (Student, Recruiter, or Admin) to log in with email and password and issue a session token on success.	Functional	High	Student / Recruiter / Admin	AC-OIPP-F-002: Valid credentials return a session token; invalid ones are rejected. Test: TC-Auth-02	None
OIPP-F-003	The system shall lock an account after 5 consecutive failed login attempts and prevent further login attempts for a defined lockout period.	Functional	Medium	Security	AC-OIPP-F-003: Account is locked after 5 failures and login is rejected during the lockout period. Test: TC-Auth-03	None

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-020	The system shall allow a prospective recruiter to submit an account request (name, company, work email) that remains in a Pending state, with no login access, until approved by the placement-cell admin.	Functional	High	Recruiter	AC-OIPP-F-020: A pending request appears in the admin's approval queue; the recruiter cannot log in until approved. Test: TC-Auth-04	Feeds into OIPP-F-018 (admin approval)

4.2 Student Profile & Resume Management

Description: Allow students to maintain a profile and upload/download a resume document.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-004	The system shall allow a student to enter academic, skill, and project details into a structured profile form.	Functional	High	Student	AC-OIPP-F-004: Profile fields save and persist across sessions. Test: TC-Prof-01	None
OIPP-F-005	The system shall allow a student to upload a	Functional	High	Student	AC-OIPP-F-005: Uploaded file is stored and	File size limit enforced

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	resume file (PDF/DOC) and replace it with a newer version.				downloadable; re-upload replaces the old file. Test: TC-Prof-02	
OIPP-F-006	The system shall generate a downloadable PDF resume from the profile data using a fixed template.	Functional	Medium	Student	AC-OIPP-F-006: Generated PDF reflects current profile data. Test: TC-Prof-03	None

4.3 Internship/Placement Listings & Search

Description: Allow students to browse and filter internship/placement listings.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-007	The system shall display a list of all active internship/placement postings to logged-in students.	Functional	High	Student	AC-OIPP-F-007: Active listings appear; closed listings are hidden. Test: TC-List-01	None
OIPP-F-008	The system shall allow a student to filter listings by role, location, and minimum CGPA eligibility.	Functional	Medium	Student	AC-OIPP-F-008: Filtered results match the selected criteria. Test: TC-List-02	None

4.4 Application & Tracking

Description: Allow students to apply to listings using their saved profile/resume and track application status.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-010	The system shall allow a student to apply to a listing using their saved profile and resume, and shall prevent duplicate applications to the same listing.	Functional	High	Student	AC-OIPP-F-010: A second application to the same listing is rejected with a message. Test: TC-App-01	Depends on OIPP-F-004/005
OIPP-F-011	The system shall show each student a status tracker for every application (Applied, Shortlisted, Interview, Selected, Rejected).	Functional	High	Student	AC-OIPP-F-011: Status updates made by recruiter/admin reflect on the student's tracker. Test: TC-App-02	Depends on OIPP-F-017

4.5 Mock Tests & Skill Assessment

Description: Provide timed multiple-choice mock tests with automatic scoring and a basic performance view.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-012	The system shall allow a student to attempt a timed	Functional	High	Student	AC-OIPP-F-012: Score is calculated and displayed	None

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	multiple-choice mock test drawn from a question bank and auto-calculate the score on submission.				immediately after submission. Test: TC-Assess-01	
OIPP-F-013	The system shall display a student's score history across all attempted mock tests.	Functional	Low	Student	AC-OIPP-F-013: Past attempts and scores are listed in chronological order. Test: TC-Assess-02	Depends on OIPP-F-012

4.6 Interview Preparation Module

Description: Provide an interview-question bank and a simple mock-interview slot booking with text feedback.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-014	The system shall provide a searchable bank of interview questions categorised by company/domain.	Functional	Medium	Student	AC-OIPP-F-014: Search/filter by category returns matching questions. Test: TC-Interview-01	None
OIPP-F-015	The system shall allow a student to book an available mock-interview slot and allow the placement-cell	Functional	Low	Student / Placement Cell Admin	AC-OIPP-F-015: The booked slot appears in the student's	No separate mentor role or calendar integration; mock-interview slots

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	administrator to record text feedback after the session.				booking list, and feedback is saved against the booking. Test: TC-Interview-02	and feedback are managed by the Placement Cell Admin

4.7 Recruiter Dashboard & Shortlisting

Description: Allow recruiters to post listings (after admin approval) and manage applicants.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-016	The system shall allow an approved recruiter to create and publish an internship/placement listing.	Functional	High	Recruiter	AC-OIPP-F-016: Listing appears to students only after recruiter approval by admin. Test: TC-Recruit-01	Depends on OIPP-F-018
OIPP-F-017	The system shall allow a recruiter to view applicants for their own listing and mark each as Shortlisted or Rejected.	Functional	High	Recruiter	AC-OIPP-F-017: Status change is reflected on the student's application tracker. Test: TC-Recruit-02	Depends on OIPP-F-010

4.8 Admin & Reporting

Description: Allow the placement-cell admin to approve pending recruiter account requests and view basic placement statistics. Admin accounts themselves are pre-created by the system/institution during setup and are not created through this module.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OIPP-F-018	The system shall allow the placement-cell admin to approve or reject a pending recruiter account request (see OIPP-F-020) before that recruiter can log in or post listings.	Functional	High	Admin	AC-OIPP-F-018: A rejected request keeps the recruiter unable to log in; an approved request unlocks login and listing creation. Test: TC-Admin-01	Depends on OIPP-F-020
OIPP-F-019	The system shall allow the admin to generate a placement summary report (applications, offers, selection rate) filterable by branch and date range.	Functional	Medium	Admin	AC-OIPP-F-019: Report totals match underlying application/offer records for the selected filter. Test: TC-Admin-02	Depends on OIPP-F-011/017

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans, and are scaled to a college mini-project deployment. IDs OIPP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OIPP-NF-001	Page load and search results shall render within 3 seconds for 90% of requests under normal (mini-project scale) load.	Performance	High	90th percentile $\leq$ 3s in load test. Test: TC-Perf-01
OIPP-NF-002	The system shall be available during declared demo/evaluation hours with no unplanned downtime.	Reliability	Medium	No unplanned outage logged during evaluation window. Test: Ops log review
OIPP-NF-003	Passwords and personal student data shall be stored securely (hashed passwords; encrypted connections).	Security/Compliance	High	Security review checklist pass. Test: TC-Sec-01
OIPP-NF-004	The system shall support at least 100 concurrent users, consistent with a single-institution mini-project deployment.	Scalability	Medium	Load test sustains 100 concurrent users at acceptable latency. Test: TC-Perf-02
OIPP-NF-005	Key student-facing workflows (registration, apply, mock test) shall be usable with a screen reader and adequate colour contrast.	Usability/Accessibility	Medium	Manual accessibility walkthrough pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Protect student and recruiter personal data from unauthorised access or disclosure.
- Ensure only authenticated, authorised users can view or modify application and shortlisting data.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OIPP-SR-001	HTTPS (TLS) shall be mandatory for all client-server connections.	Security	High	AC-OIPP-SR-001: HTTP requests are redirected to HTTPS. Test: TC-Sec-02
OIPP-SR-002	The system shall enforce role-based access control (Student, Recruiter, Admin).	Security	High	AC-OIPP-SR-002: Actions outside a role's permissions are rejected. Test: TC-Sec-03
OIPP-SR-003	The system shall store passwords using a salted, industry-standard hashing algorithm.	Security	High	AC-OIPP-SR-003: Password hash review passes review. Test: TC-Sec-04
OIPP-SR-004	The system shall log security-relevant events (login, role change) with a timestamp for the duration of the project evaluation period.	Security	Medium	AC-OIPP-SR-004: Sample audit log entries verified complete. Test: TC-Sec-05
OIPP-SR-005	The system shall issue session tokens (JWT) with a defined expiry, requiring re-login after expiry.	Security	Medium	AC-OIPP-SR-005: Expired tokens are rejected by the API. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.
- Acceptance test suites: Registration & Authentication, Profile & Resume, Listings & Application, Mock Tests, Interview Prep, Recruiter Dashboard, Admin & Reporting, Performance, Security, and Accessibility tests.

7. System models and diagrams

7.1 UML Use-Case diagrams

Figure 1 — Student use cases: Register, Log In, Build/Update Profile, Upload Resume, Search Listings, Apply to Listing, Track Application Status, Take Mock Test, Book Mock Interview.

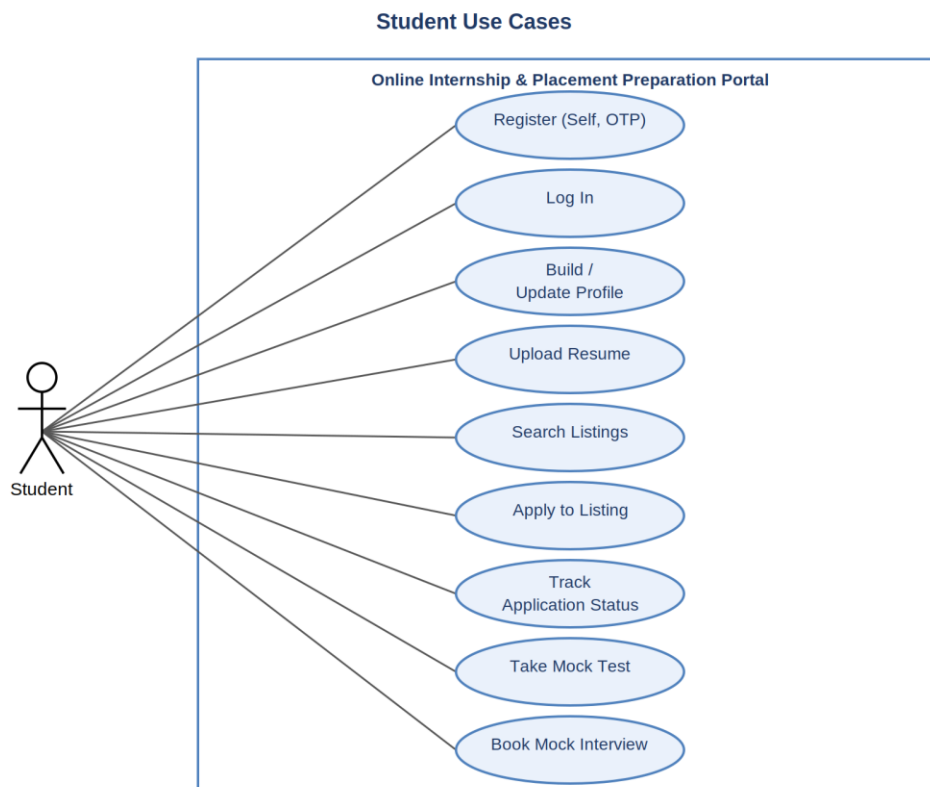
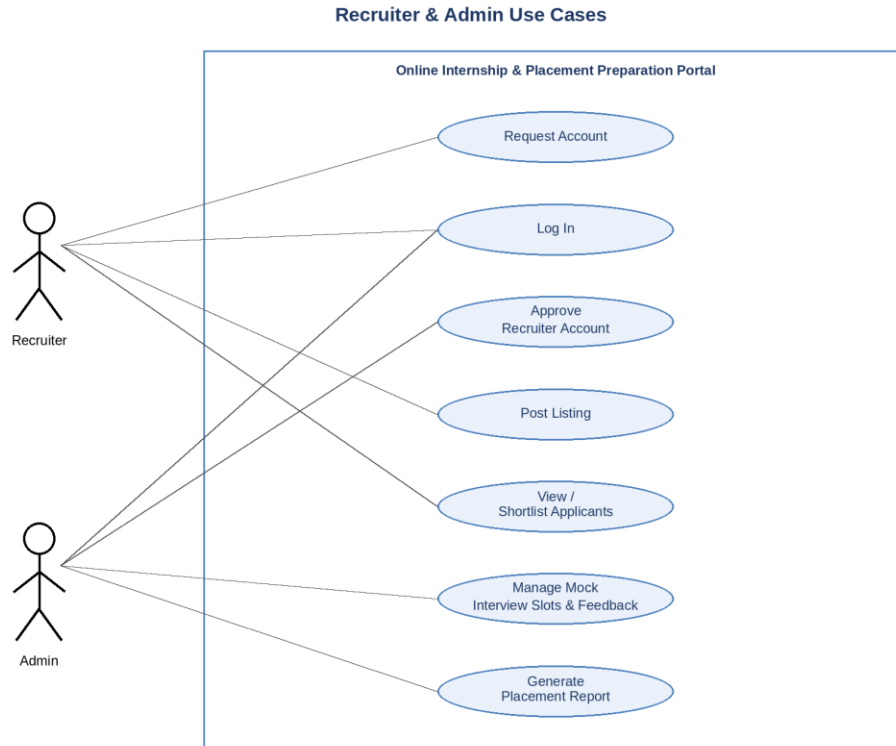


Figure 2 — Recruiter/Admin use cases: Request Account (Recruiter), Log In (shared), Approve Recruiter Account (Admin), Post Listing (Recruiter), View/Shortlist Applicants (Recruiter), Manage Mock Interview Slots & Feedback (Admin), Generate Placement Report (Admin).



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OIPP-F-001	Register with OTP	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
OIPP-F-002	Login & session token	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	
OIPP-F-003	Lockout after 5 failures	4.1 / DS-Auth-03	AuthModule	TC-Auth-03	N	
OIPP-F-020	Recruiter account request	4.1 / DS-Auth-04	AuthModule	TC-Auth-04	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OIPP-F-004	Structured profile form	4.2 / DS-Profile-01	ProfileModule	TC-Prof-01	N	
OIPP-F-005	Resume upload/replace	4.2 / DS-Profile-02	ProfileModule	TC-Prof-02	N	
OIPP-F-006	Generate PDF resume	4.2 / DS-Profile-03	ProfileModule	TC-Prof-03	N	
OIPP-F-007	List active postings	4.3 / DS-Listing-01	ListingModule	TC-List-01	N	
OIPP-F-008	Filter listings	4.3 / DS-Listing-02	ListingModule	TC-List-02	N	
OIPP-F-010	Apply, prevent duplicates	4.4 / DS-App-01	ApplicationModule	TC-App-01	N	
OIPP-F-011	Application status tracker	4.4 / DS-App-02	ApplicationModule	TC-App-02	N	
OIPP-F-012	Mock test auto-scoring	4.5 / DS-Assess-01	AssessmentModule	TC-Assess-01	N	
OIPP-F-013	Score history view	4.5 / DS-Assess-02	AssessmentModule	TC-Assess-02	N	
OIPP-F-014	Interview question bank	4.6 / DS-Interview-01	InterviewPrepModule	TC-Interview-01	N	
OIPP-F-015	Mock-interview booking	4.6 / DS-Interview-02	InterviewPrepModule	TC-Interview-02	N	
OIPP-F-016	Recruiter posts listing	4.7 / DS-Recruit-01	RecruiterModule	TC-Recruit-01	N	
OIPP-F-017	Shortlist applicants	4.7 / DS-Recruit-02	RecruiterModule	TC-Recruit-02	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OIPP-F-018	Approve recruiter	4.8 / DS-Admin-01	AdminModule	TC-Admin-01	N	
OIPP-F-019	Placement summary report	4.8 / DS-Admin-02	AdminModule	TC-Admin-02	N	
OIPP-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
OIPP-NF-002	Availability during evaluation	5 / DS-Ops-01	Infra	Ops log review	N	
OIPP-NF-003	Secure data storage	5 / DS-Sec-01	Infra	TC-Sec-01	N	
OIPP-NF-004	100 concurrent users	5 / DS-Perf-02	WebUI / API	TC-Perf-02	N	
OIPP-NF-005	Accessibility of key flows	5 / DS-UX-01	WebUI	TC-UX-01	N	
OIPP-SR-001	HTTPS enforcement	5.1.2 / DS-Sec-02	Infra	TC-Sec-02	N	
OIPP-SR-002	Role-based access control	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03	N	
OIPP-SR-003	Password hashing	5.1.2 / DS-Sec-04	AuthModule	TC-Sec-04	N	
OIPP-SR-004	Security event logging	5.1.2 / DS-Sec-05	Infra	TC-Sec-05	N	
OIPP-SR-005	JWT session expiry	5.1.2 / DS-Sec-06	AuthModule	TC-Sec-06	N	